

The Cost of Social Norms: Helping Behavior in the Field

A Field Experiment on Social vs. Market Norms

Presented by : Zest (Group 8)

Pragya, Kashpreet, Anisha,
Aayushi, Radhika, Tango



Overview

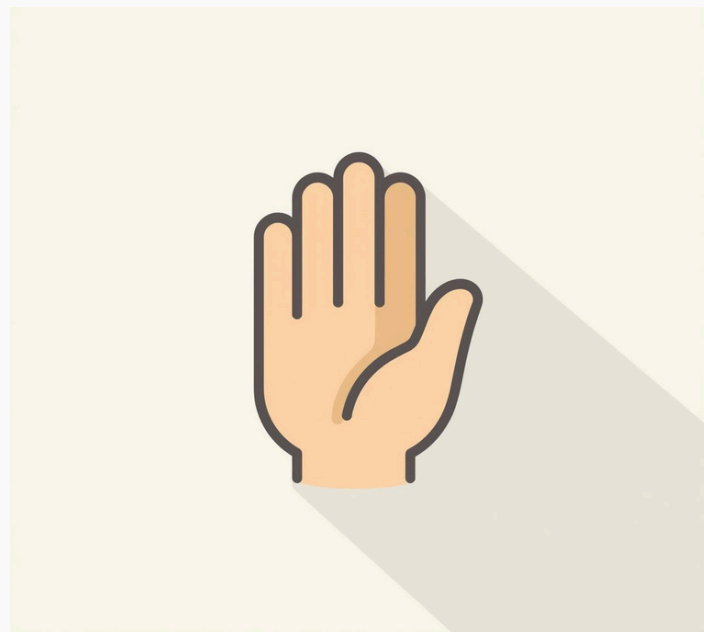
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Hypothesis Identification and Formulation

Individuals are likely to be more helpful to others when no reward is offered (social norms) than when a reward is offered (market norms). Under market norms, individuals are likely to be more helpful when an incentive in kind (e.g., a candy bar) is offered than when a monetary incentive (e.g., 50 cents) is offered.

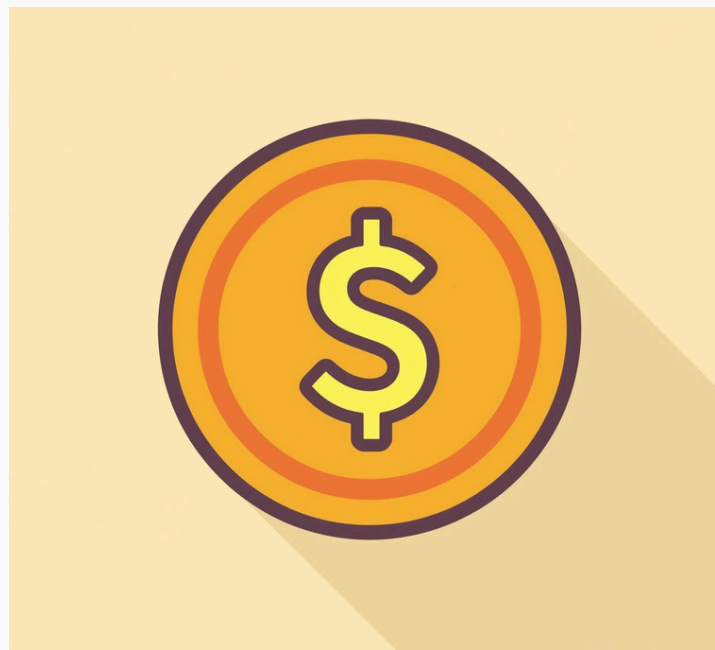
No Reward

Participants receive no incentive for helping.



Monetary Reward

\$0.50 incentive for helpful behavior.



Candy Bar

Token gift to encourage prosocial actions.



Hypothesis Identification and Formulation

Reward framing (no reward vs. \$0.50 vs. candy bar) is likely to influence helpfulness, operationalized as a 1–5 helpfulness score. Average helpfulness scores are expected to be highest in the No Reward condition, followed by Candy Bar condition, and lowest in the \$0.50 condition.

3 Conditions

Types of reward framing

5-Point Scale

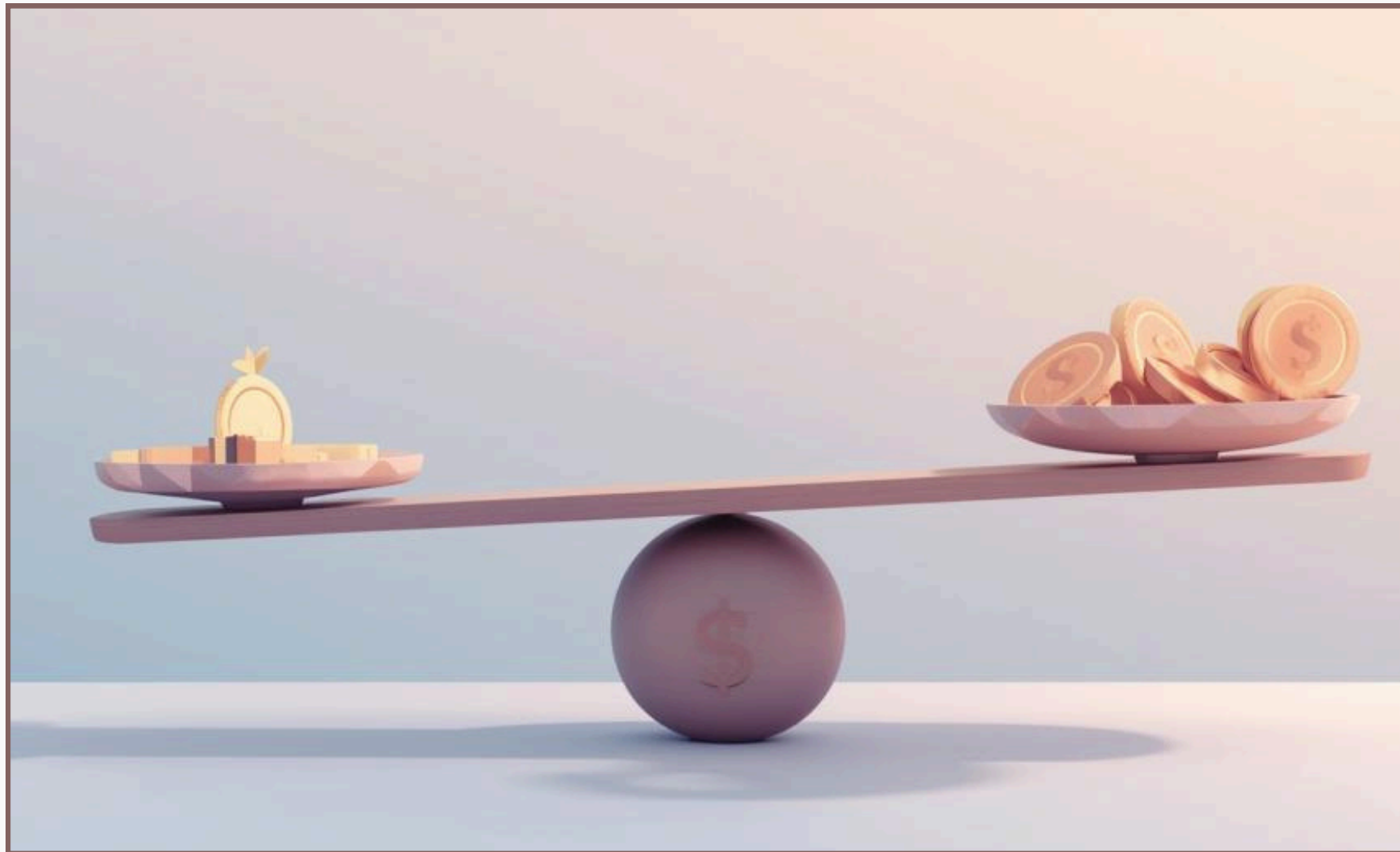
Helpfulness rating system

15 Participants

Size of each experimental group

Identification of Experimental Variables

Understanding Key Elements of the Study



Independent Variable

Type of Reward

- No Reward
- \$0.50 Monetary Reward
- Candy Bar Token

Dependent Variable

Helpfulness Score Scale from 1 to 5

- 1 = Completely unhelpful
- 2 = Less helpful
- 3 = Fair / neutral
- 4 = Helpful
- 5 = Very helpful

Identification of Experimental Variables

Measured Variables

- Helpfulness score (1–5)
- Whether the participant physically walked us to the aisle
- Whether they pointed vs. verbally guided
- Whether the participant accepted or refused the reward (in reward conditions)
- Qualitative comments made by participants



Controlled Variables

- One primary experimenter consistently asked the questions across all conditions.
- A second experimenter remained present to support and ensure consistency if anything was missed.
- The wording of the request was standardized, with only the reward phrase changing.
- The same task (finding dishwasher pods) was used for all participants.
- All interactions took place inside the same Costco location.
- The same three teammates served as observers throughout the study.
- Observers recorded notes consistently via WhatsApp.
- We always acted as if we followed the participant's directions.
- We moved throughout the store rather than staying in one area.
- One experimenter carried a shopping trolley during all interactions.
- The same 1–5 helpfulness scale was used for every participant.

Experimental Design, Method and Groups



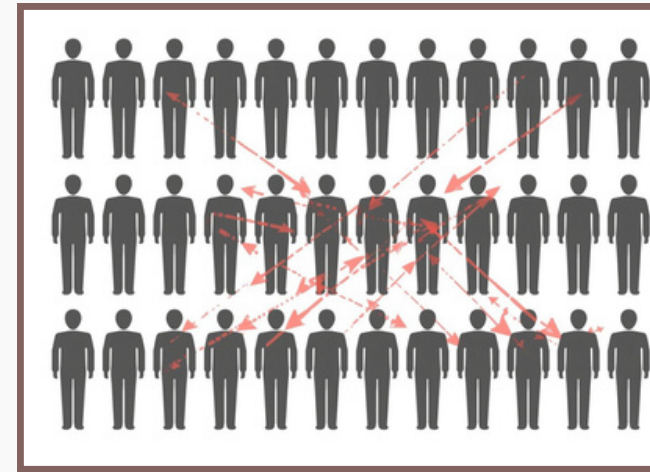
Manipulation

Active manipulation of independent variables was applied.



Random Assignment

Attempted random assignment ensured balanced participant groups.



Controlled Factors

Key factors were kept consistent throughout the experiment.



Supportive Procedures

Dual experimenters provided support during the study.



Reliability

Group consensus on ratings ensured data reliability.

Experimental Design, Method and Groups

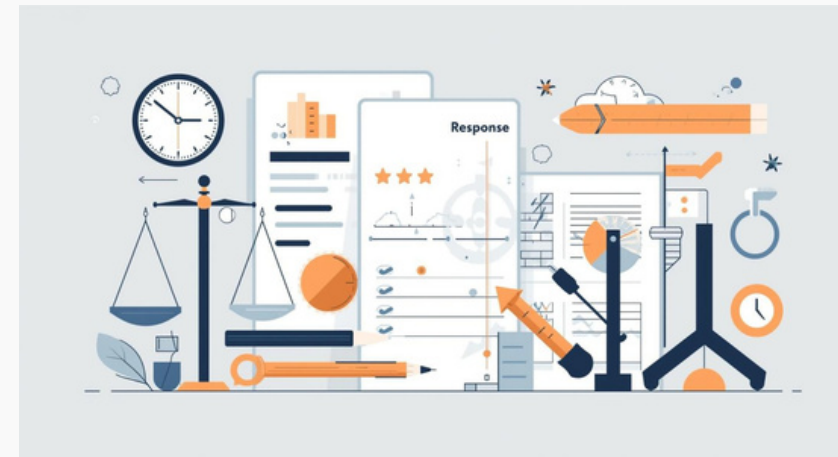
01



Standardized Script

Ensured consistency in communication across participants.

02



Consistent Measurement

Ensured reliable and objective data collection methods.

03



Consistent Experimenters

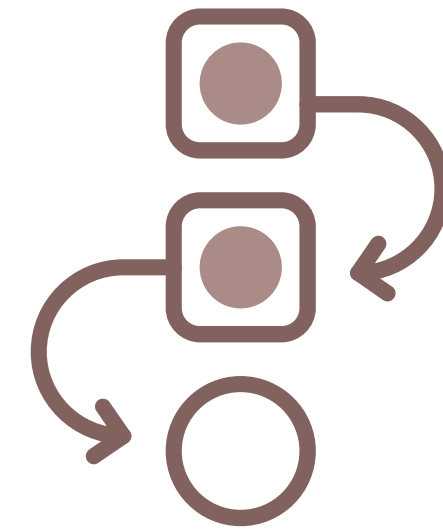
Maintained uniform delivery of procedures and instructions.

Causality Establishment



Temporal Precedence

- Temporal precedence established by presenting the reward framing (IV) before any helping behavior (DV) was measured
- Participants first heard one of three request versions: No reward, 50 cents, or Candy bar
- Helpfulness was rated after the request on a 1–5 scale
- This sequencing ensures the cause (reward framing) occurred before the effect (helpfulness)



Covariation

- Covariation means that changes in the reward framing corresponded to changes in helpfulness.
- With three equal-sized groups (No reward, 50 cents, Candy bar), any differences in behavior can be tied to the manipulation rather than pre-existing differences.
- The results showed a clear pattern: the monetary reward produced the lowest helpfulness, the candy bar the highest, and no reward fell in between.
- This variation demonstrates that shifts in the independent variable were directly associated with shifts in the dependent variable, satisfying the covariation criterion for causal inference.

Causality Establishment



Elimination of Alternatives

- Held all non-reward factors constant to rule out alternative explanations
- Same experimenter and standardized script across all conditions
- Single setting (same Costco) and identical task (locating dishwasher pods)
- Participants approached without demographic targeting; screened out those visibly rushed or distracted
- Movement throughout the store to avoid location-specific bias
- Immediate post-interaction helpfulness ratings with group consensus to reduce rater bias
- Equal group sizes (15 per condition) to maintain balance

Validity and Confounding Variables

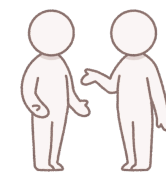
Internal Validity: Low

Independent Variable produced an effect on the dependent variable but it was not in the direction of our hypothesis. We held several important varying conditions constant.

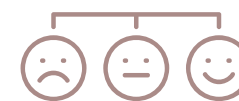
Threats



Situational effects - Conducting experiment at different times



Experimenter effects - Unintentional changes in body language/tone when offering money vs. candy.



Measurement Bias - Subjectivity in the 1–5 rating scale without multiple blind raters.



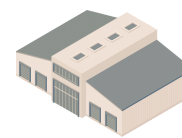
Hypothesis Guessing: Participants may have guessed the study's purpose, especially when offered cash.

Validity and Confounding Variables

External Validity: Low

The findings may not generalize to other populations, settings, or higher-effort helping situations. Therefore, while the experiment provides useful insight into behavior within this specific context, the results should not be assumed to hold universally across people or environments.

Threats



Setting Specificity: Costco is unique, large-format store, high foot traffic.



Population Specificity: Participants likely not representative of all consumers (age mix, language, culture, time pressure).



Time window: Results may not generalize to weekdays or other seasons.



Task specificity : Effects might differ for higher-effort helping.

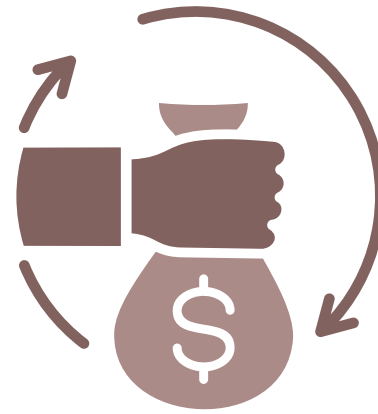
Validity and Confounding Variables

Confounding Variables



Time of day

Fluctuations in store business could introduce time pressure as a hidden variable.



Script Delivery

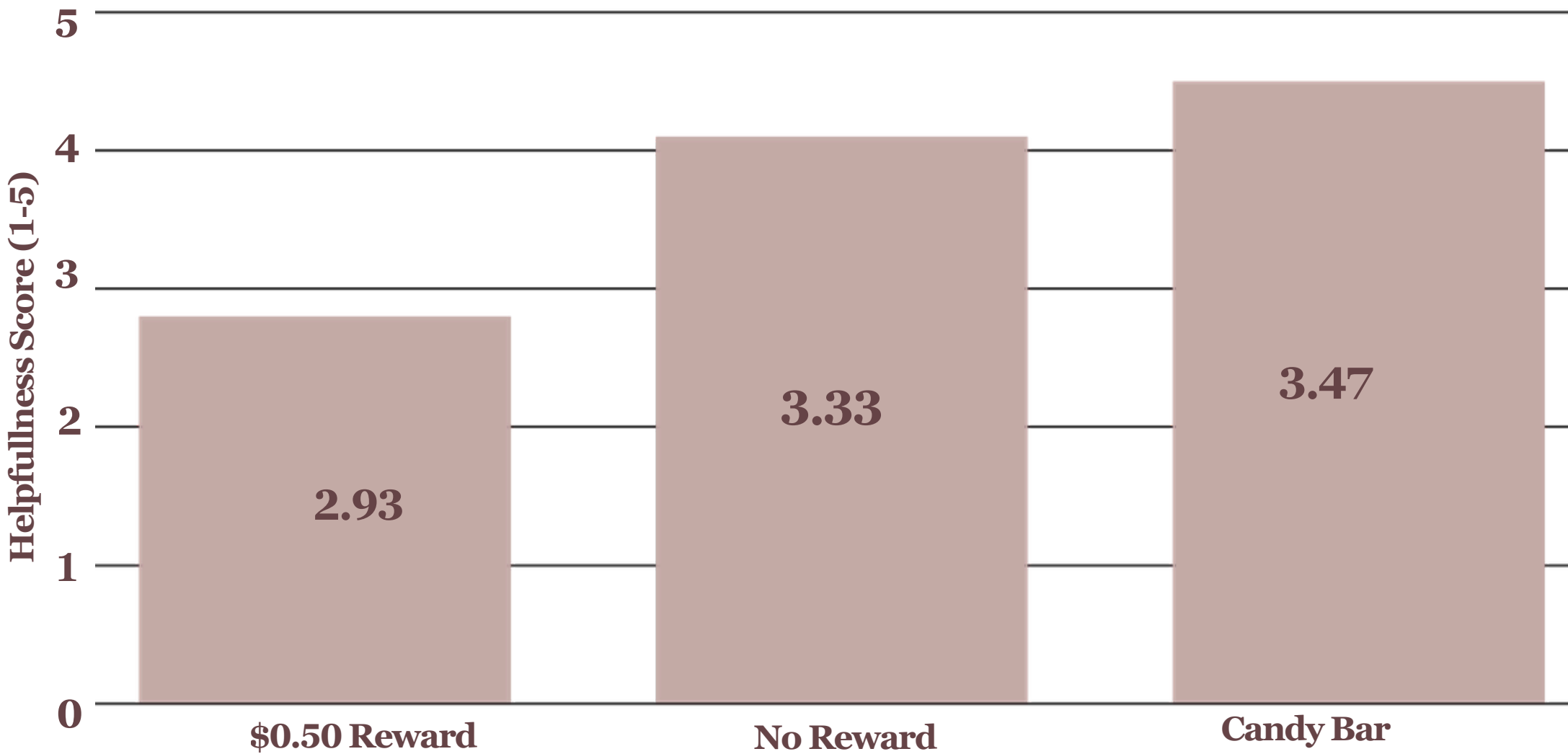
Offering money can feel "transactional," leading to awkwardness that reduces helpfulness independent of the reward value.



Order Effects

Running conditions in a fixed sequence, fatigue or learning could affect later conditions.

Data Analysis and Hypothesis Support



Key Observations

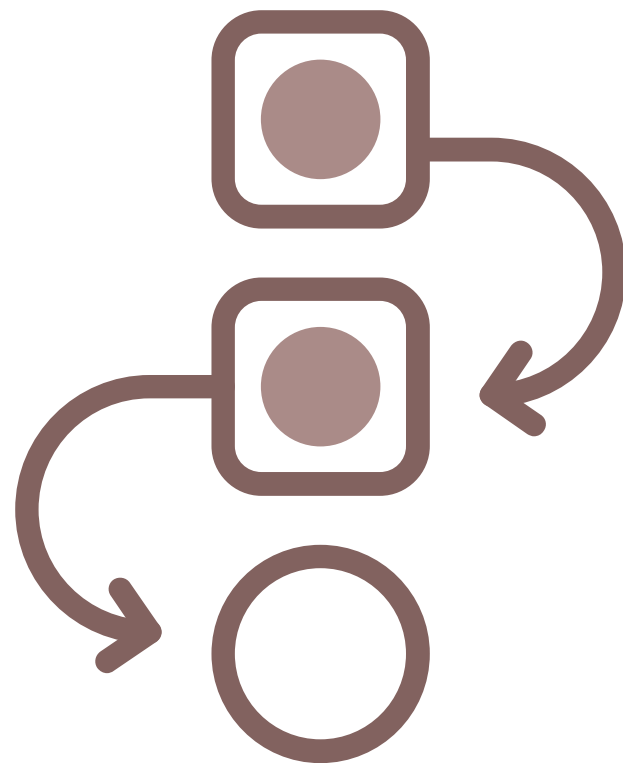
Highest Helpfulness: The Candy Bar condition acted as a strong social cue, exceeding the baseline.

Lowest Helpfulness: The \$0.50 Reward yielded the lowest scores, confirming the "crowding out" of social motivation.

Baseline: No Reward showed moderate to high helpfulness, typical of social norm interactions.

Data Analysis and Hypothesis Support

01 The findings partially support our hypothesis that social-norm framing (gift or no reward) influences participant helpfulness



However, Candy slightly exceeded No Reward, suggesting a small gift served as a stronger social cue in this setting.

Covariation

Changing the IV (reward framing) produced systematic differences in the DV (helpfulness ratings) across conditions

Clear pattern in means

Candy highest, No Reward next, \$0.50 lowest—even if not perfectly matching the predicted order, it still shows the IV matters.

Mechanism (why it happens)

Introducing money can shift the interaction into a market frame, which can crowd out social motivation, while a small gift/free can activate a social frame.

Data Analysis and Hypothesis Support

Small cash incentives can backfire relative to no reward, while small gifts can sustain or increase helping, making the result actionable for real-world settings

Why hypothesis testing matters

The study tests whether framing shifts behavior via social vs. market norms, not just whether “people are helpful at Costco.”

Decision value of comparisons

Using multiple comparison groups (treatments) lets us interpret differences across conditions and make clear recommendations based on what actually changed outcomes (not gut feeling)

Real-world strength

Because this was a field setting, the observed effects reflect how people respond to incentives in natural, everyday interactions, making the insights more practical and generalizable than a purely lab-based test

Hypothesis Guessing and Demand Artifacts

01

Why it was likely low here?

The interaction looked like a normal shopper request (quick ask for help), not an obvious “study,” so there were fewer cues to guess the hypothesis.

02

Single-shot exposure

Each participant experienced only one condition once (no repeated tasks), so it’s harder to notice a pattern or infer that rewards are being tested.

03

Observer camouflage

Data recording was done discreetly (observers physically distant / not in direct line of sight), reducing reactivity from being watched.



04

Condition separation

The team regrouped only after completing the full condition and discussed privately, minimizing visible “research team” behaviors that could tip people off.

05

Most plausible remaining artifact (honest limitation)

The \$0.50 offer can feel unusual/transactional, which might create mild suspicion or discomfort—so part of the lower helpfulness could reflect reactance/awkwardness, not only the intended mechanism.

Alternative Experimental Design and Study Improvement

01

True individual-level random assignment

Participants were assigned to conditions sequentially.

Future studies should randomly assign each participant to a condition.

This would reduce time and environment differences across conditions and strengthen causal conclusions (Mittal, 2016).

02

More objective and multi-dimensional measurement of helpfulness

Helpfulness was measured using a 1–5 rating scale, which included some subjectivity.

Future studies could use a structured checklist:

- Walked to aisle
- Gave detailed directions
- Pointed only

This would improve measurement reliability and precision.



Alternative Experimental Design and Study Improvement

03

Stronger manipulation testing

Future studies could test different reward levels:

\$0.50 vs \$2 vs \$5

Different types of gifts

This would help determine whether behavior changes because of money itself or reward size.

04

Include additional dependent variables beyond helpfulness

Future studies could measure:

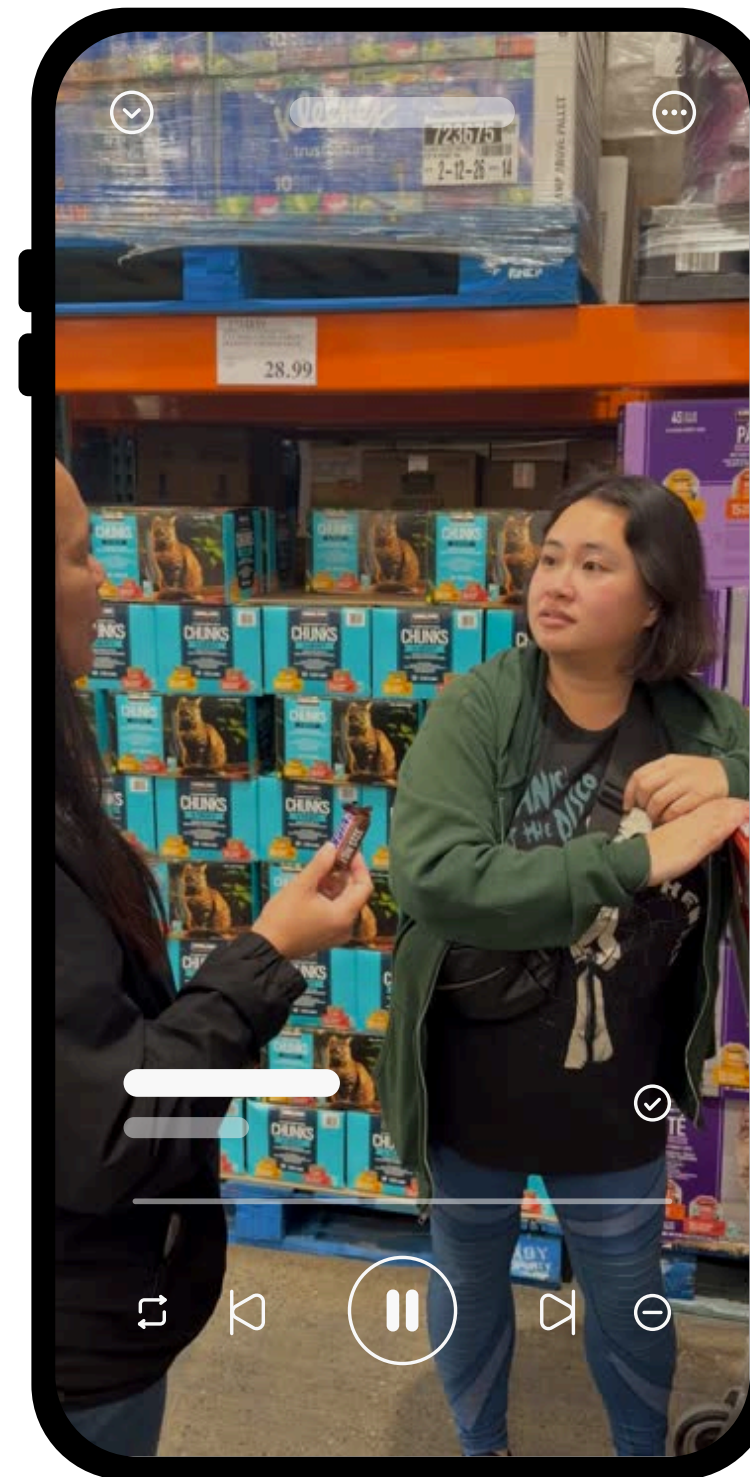
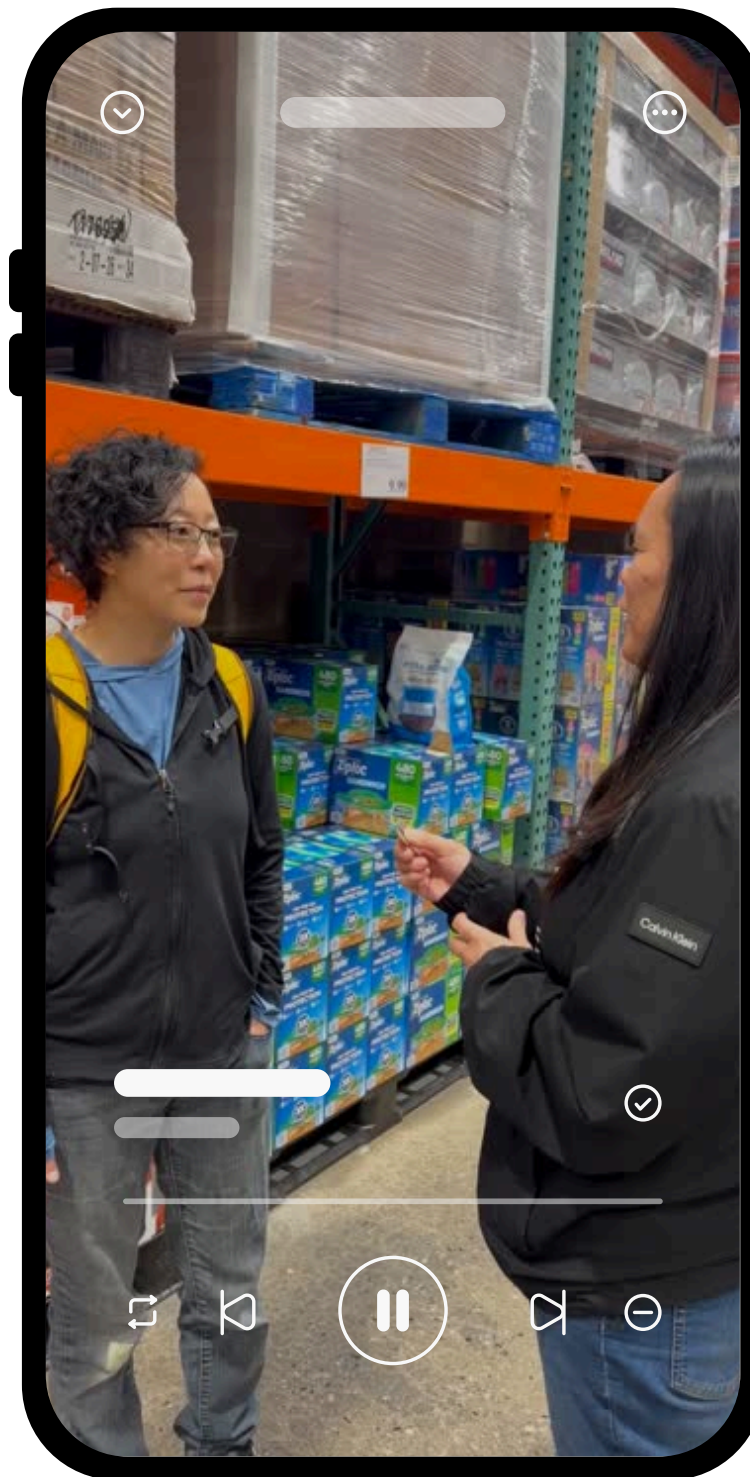
Emotional reactions

- Response time
- Verbal tone

These measures could better capture differences between social norms and market norms.



Behind the Scenes!



Thank You

